Storm Night: who sails first when the ships don’t?

IT business analysis portfolio case · Philippe Godfroy · fictional RoRo operator (Tidewell Shortsea Lines), illustrative figures, no employer data

The situation. Thursday 16:00, a storm cancels three sailings and strands 859 trailers. Today rebooking waits for customers to call: perishables miss their deadline while frozen cargo sails, the yard overflows, and contract customers hear from their account manager the next morning. Sales, customer service and operations each asked for a different fix.

The reframe. Not “rebook faster” but *time to certainty*: every customer gets a new sailing, with a reason, within the hour, in an order all three departments can defend. A duty manager approves a rule-based recovery plan instead of waiting for phone calls.

RESULT · MEDIAN OF 200 SIMULATED NIGHTS

	AS-IS	TO-BE
Hours until 90% of stranded units have a confirmed sailing	17.1	**0.8**
Perishables sailing after their deadline	21	**10**
Hours the yard is over capacity	12.7	**0**
Hours drivers wait at the terminal	3,361	**1,841**
Units withdrawn to a competitor	54	**1**
Calls to customer service	116	**20**
Average delay	±19 h	±20 h

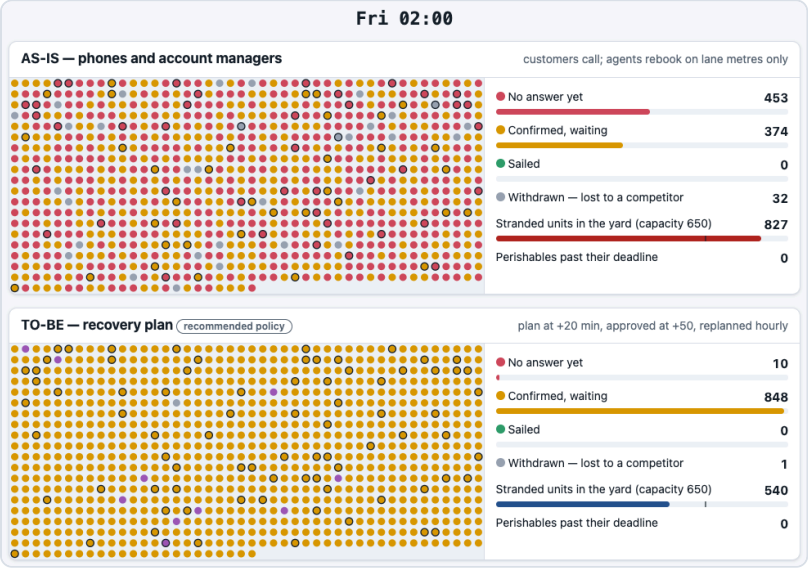
The average delay does not change: a storm removes capacity and no rule creates a ship. The business case says so.

BUSINESS CASE, AS A RANGE

€763k 5-year NPV (P10 €382k, P90 €1,086k) · payback 13 months · **6.5%** chance of a loss when all 44 assumptions are uncertain at once. A tornado chart names the three assumptions to verify first.

WHAT I DID AS THE ANALYST

- Reconstructed the last storm night from data before the first interview
- 12 stakeholders, 18 voices; kept the contradiction (Sales thought key accounts came first; the data said last)
- Event storming, BPMN 2.0 AS-IS / TO-BE, UML sequence, state and data model
- 16 allocation rules with 23 executable Given/When/Then examples
- Policy workshop: four proposals simulated on the same nights; the simulation exposed a flaw in Operations’ proposal (→ rule R-15)
- 17 user stories, WSJF prioritisation, five screens validated in four rounds
- Business acceptance run as a storm drill; it found a severity-1 gap in my own rules (→ R-16)
- Pre-mortem, shadow mode, go/no-go, rollback, hypercare and a benefits review



The same storm, replayed twice, at Friday 02:00. One dot per stranded unit: red = no answer yet, amber = new sailing confirmed, grey = withdrawn, ringed = perishable. Top: how it works today, where customers call and wait. Bottom: the recovery plan, built 20 minutes after the cancellation and approved by the duty manager at +50.

WHY IT CAN BE TRUSTED

Every number above is regenerated in CI from 44 stated assumptions; a test fails if a document quotes a number the model does not produce. A traceability check fails the build on a pain nobody voiced, a rule nobody tested or an ID that does not exist. Built by one person with AI assistance; every decision and its reasoning is in the decision log.

Replay the night (2-minute tour): phlppgdfry.github.io/storm-night-disruption-business-analysis/war-room/?tour
Case and documents: github.com/phlppgdfry/storm-night-disruption-business-analysis